MITTAG-LEFFLER IN THE FIRST TERM MAKES A SEQUENCE OF LIMITS EXACT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file homology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; every system it uses is displayed and each application of the cited lemma is justified by a property established in the preceding line, and no step is left to the reader.

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Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the comment-type reference, slogan and history environments (native preamble lines 8-13), the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217, the actual last line of the file).

Allowed standard prerequisites (editorial).

- The Mittag-Leffler condition and limits. Same chapter, Lemma lemma-Mittag-Leffler. Supplies that a quotient of a Mittag-Leffler system is Mittag-Leffler and that the limit of a short exact sequence of inverse systems is exact when the left term is Mittag-Leffler, which is applied twice.
- Limits commute with kernels. Same chapter, the standard identification of the limit of the kernels with the kernel of the limit. Supplies the description of the limit of the kernel systems appearing in the proof.

Source reference. Original lemma 02N1: sources/stacks-homology/pinned-original-source.tex, source lines 7890–7890. Navigation pointer invoked at proof line 7938; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let*

$$(A_i) \to (B_i) \to (C_i) \to (D_i)$$

be an exact sequence of inverse systems of abelian groups. If the system (A_i) is ML, then the sequence

$$\lim_i B_i \to \lim_i C_i \to \lim_i D_i$$

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is exact.

Proof. Let $Z_i = \mathrm{Ker}(C_i \to D_i)$ and $I_i = \mathrm{Im}(A_i \to B_i)$. Then $\lim Z_i = \mathrm{Ker}(\lim C_i \to \lim D_i)$ and we get a short exact sequence of systems

$$0 \to (I_i) \to (B_i) \to (Z_i) \to 0$$

Moreover, by Lemma 02N1 we see that (I_i) has (ML), thus another application of Lemma 02N1 shows that $\lim B_i \to \lim Z_i$ is surjective which proves the lemma. $\square$